Makers Making Change
A Neil Squire Program

Neil Squire

Toy Adapting Instructions

None

None

None

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1. Latest Updates

Version 1.0 | July, 2026 | Original version of this content

Creation of this website. The goal was to mimic the functionality of the GitHub repo but with better UI and easier searching for the toy instructions. This then will be evaluated to see how it can be improved.

1.1 In Progress

A running list of resources we want to create. If you have time to help us work on these, please email us at info@makersmakingchange.com.

- Nothing at the moment.

2. Welcome to MMC's Toy Adapting Instructions

2.1 What is this Resource?

Welcome! This resource is a collection of the Makers Making Change (https://www.makersmakingchange.com/), a program of Neil Squire (https://www.neilsquire.ca/), toy adapting instructions and resources. Intended for makers and families to find step by step instructions to adapt toys to be assisitive switch accessible and host resources to make that process easier.

2.2 Your Pathway to Play

There are two main sections of this web page:

2.2.1 Toy Adapting Instructions

A variety of toys ranging in play styles (bubbles, light and sound, etc.) with step by step instructions on how to adapt them to be switch accessible.

- Check out the toy instructions here.

2.2.2 Toy Adapting Resources

Resources to help learn about toy adapting.

- Check out the toy resources here.

2.3 Hacking For the Holidays

We need your help to make play accessible for kids with disabilities. Every year from September to December, volunteers across Canada join a mission to adapt and donate accessible toys and switches to families and clinicians nationwide.

Join us for the 2026 Hacking for the Holidays, presented by Air Canada Foundation, as we fundraise $150,000, engage 1,000 volunteers, and deliver 2,500 devices. Learn more on the Hacking For the Holidays site. (https://www.makersmakingchange.com/hacking-for-the-holidays)

2.4 Why Adaptive Toys Matter

For many kids with disabilities, toys can be hard to use independently, and commercially adapted versions can cost hundreds of dollars. That high cost leaves many families and therapists supporting kids with disabilities without the interactive toys and access methods crucial to childhood development. However, with a little bit of tinkering, toys can be switch-adapt and made accessible for a fraction of the cost.

3. Toy Instructions

Browse switch-adapted toy builds. Use the filters below to narrow down by category, or scroll through everything.

Some info may be out of date

This resource is where Makers Making Change hosts all of our toy instructions both current and past. This means, some of the toys that you find instructions for may not be available at the moment OR information may be out of date.

Filters ▾

Clear Filters

4. Component & Tool List

This is a list of commonly used components and tools for switch adapting toys. Most components can be purchased from different manufacturers for various prices, and the tools listed are widely available at most local hardware stores but you can also use the link below to find our recommended links. Not every tool and component is needed for every toy. Please check the first page of a toy's Maker Guide to see what it specifically calls for.

4.1 Components

4.1.1 Mono Jack

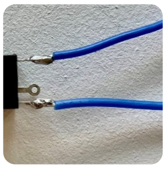
3.5 mm Rectangular Mono Jack

The most used mono jack. Solder wires onto the two tabs closest to the cable input.

Example: wires soldered onto the two tabs closest to the cable input

3.5 mm Square Mono Jack

Solder wires onto the two tabs directly across from each other.



Example: wires soldered onto the two tabs directly across from each other

3.5 mm Mono Jack Flat Rectangular

3.5 mm Flat Mono Jack

4.1.2 Cables

3.5 mm F/Bare Mono Cable (1 ft)

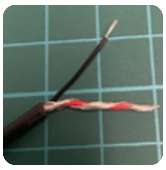
~$20.22 CAD / 10-pack

Standard mono cable, ready to use as-is.

3.5 mm F/Bare Stereo Cable (1 ft)

~$10.99 CAD / 10-pack

To use as a mono cable: fully strip the insulation off the white wire, and 0.5 cm from the red and black wires. Wrap the exposed wire around the red cable.



Example: stripped and wrapped to use as a mono cable

4.1.3 Battery Interrupter

Flex PCB

Request directly from Makers Making Change.

4.1.4 Additional Components

22 AWG Wire

Any 22 AWG silicone wire will work.

10,000 pf (103) Capacitor

Used for adapting multiple RC cars, including the Amazon Dinosaur RC Car, the Prextex RC Race and Police Car, and the Prextex RC Truck and Tractor.

Toggle Switch

~$15.98 CAD / 10-pack

Used in the Nerf N-Strike Elite Hyperfire Blaster.

4.2 Tools

Soldering Kit

~$39.99 CAD

An easy and cheap way to acquire a lot of tools at once. Includes a soldering iron, solder, soldering iron stand, sponge, Phillips and flathead screwdrivers, flush cutter, tweezers, and more.

Wire Strippers

~$21.43 CAD

Required for mono jack / cable toys.

Multi Head Screwdriver

~$31.77 CAD

Comes with 26 bits. Helpful to have a few more screwdrivers than soldering irons on hand at events. Most toys use Phillips screwdrivers.

Cable Tie

~$10.99 CAD

Attach to the mono cable on the inside of the toy for strain relief. A pack of 150 typically includes 50 each of 10cm, 20cm, and 30cm ties.

Flush Cutter

Used to cut wires. Already included in the soldering kit above.

Drill and Drill Bits